

The discourse-sensitive meaning of English *already* and its Paraguayan Guaraní counterpart¹

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1 Overview

already-like expressions have received attention in the semantic literature for their focus-sensitivity and polysemy (e.g., Löbner, 1989; Krifka, 2000; Ippolito, 2007).

- However, less is known about their semantic properties across languages (see, though, Mucha, 2015 on Medumba *yǎ*; Vander Klok & Matthewson, 2015 on Javanese *wis*).

In this talk, we introduce data on the aspectual suffix *-ma* in Paraguayan Guaraní (Tupí-Guaraní), which we identify as a counterpart to English *already*.

We establish the existence of **two types of aspectual *already*s** cross-linguistically.

- **Type A:** *Entails* a change of state.
 - Paraguayan Guaraní *-ma*, Bulgarian *veče*
- **Type B:** Only *implicates* a change of state.
 - English *already*, Spanish *ya*

Proposal: Cross-linguistically, *already*s **reference the discourse structure**, following Toosarvandani (2014) on counterexpectational *but*.

- Allows us to synthesize divergent aspects of past accounts of *already* (e.g., Löbner, 1989; Krifka, 2000; Ippolito, 2007).
- Captures both the **aspectual** and **marginal** uses of *already*, shared by English, Spanish, Paraguayan Guaraní and Bulgarian.

Data comes from fieldwork with Paraguayan Guaraní speakers, using standard semantic elicitation tasks (e.g., Matthewson, 2004).

- Consulted speakers are Spanish-Paraguayan Guaraní bilinguals.
- Additionally speak Jopara, a variety of code-switching between Spanish and Guaraní.

¹We are grateful to Agus Toñanez and Elva Gómez for sharing their knowledge of Paraguayan Guaraní with us. We also thank Eli Sharf, Josefina Bittar Prieto, Pranav Anand, Katherine Russell, Aidan Katson, and Iliyana Krapova for helpful discussions and feedback on this work, as well as judgments.

Roadmap of talk:

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§6 Conclusion

2 Paraguayan Guaraní *-ma* is a type of *already*

The semantics of Paraguayan Guaraní *-ma* has remained elusive, variably described as:

- A perfect (Tonhauser, 2006; 2011)
- A perfective (Estigarribia, 2017)
- An *already* (Gregores & Suarez, 1967; Estigarribia, 2020; Russell, 2020)
- A grammaticalized form of *already* (i.e., a ‘iamitive’ aspect) (Olsson, 2013; Dahl & Wälchli, 2016)

We argue that Paraguayan Guaraní *-ma* is an *already*, via four diagnostics from Löbner (1989) and Vander Klok & Matthewson (2015).

i. *-ma* comes with a **counterexpectational ‘earliness’ implication**, like English *already* and Spanish *ya*.

- **‘Earliness’ implication:** An eventuality *e* holds at an earlier time than anticipated (1).
- *e* can be either ongoing (1a) or completed (1b), indicating that *-ma* is not perfective.

(1) Context: *The weather forecast says it will rain tonight. It is noon now, ...*

a. ...and it’s pouring.

O-ký-**ma** (hína).

3.ACT-rain-*ma* (CONT)

‘It is **already** raining.’

b. ...and it had rained at 9am. The rain has stopped.

O-ký-**ma**.

3.ACT-rain-*ma*

‘It **already** rained.’

ii. *-ma* is in **complementary distribution with *gueteri* ‘still’**, its inverse (Löbner, 1989; Krifka, 2000; Ippolito, 2007).

- *Already*: Implicates that an eventuality *e* holds at an **earlier** time than expected.
- *Still*: Implicates that an eventuality *e* holds at a **later** time than expected.
- Cross-linguistically, *still* requires *e* to be ongoing/imperfective (e.g., Michaelis, 1993; Aonuki, 2021).

- Yet, even when *e* is ongoing at the reference time, *gueteri* ‘still’ is incompatible with *-ma*.

- (2) Luis o-dipara(**#-ma**) gueteri.
 Luis 3.ACT-run(*-ma*) still
 ‘Luis is still (**#already**) running.’

iii. *-ma* exhibits **duality when interacting with negation** (e.g., Löbner, 1989), like *already* and *ya*.

- Like how negation results in equivalences between quantifiers (e.g., $\neg\forall x[P(x)] \Leftrightarrow \exists x[\neg P(x)]$), the scope of negation relative to *already* leads to truth conditions that are equivalent to its counterparts *still* / *not anymore*.
- In both (3a) and (3b), *already*($\neg p$) is semantically equivalent to $\neg$ *still*(*p*) (i.e., *not anymore*).

- (3) a. I **already** don’t go to school.
 = ‘I don’t go to school anymore.’
 b. Nd-a-ha-vé-i-**ma** eskuéla=pe.
 NEG-1.ACT-go-more-NEG-*ma* school=LOC.
 ‘I don’t go to school anymore.’

iv. *-ma* and *already* both result in **a change-of-state inference with statives**.

- Consider the infelicity of the predicate *young* with *-ma* / *already* as compared to *old* (4).
- It is impossible to change from a preceding state of *not young* to *young* per its lexical semantics (Löbner, 1989), resulting in the oddness of (4a).
- An analysis of *-ma* as a perfect predicts felicity in (4a), as no incompatibility should arise (4c).

- (4) a. #Jagua i-mitã-**ma**.
 dog 3.INACT-young-*ma*
 Intended: ‘The dog is **already** young.’
 b. Jagua i-guaiguĩ-**ma**.
 dog 3.INACT-old-*ma*
 ‘The dog is **already** old.’
 c. The dog has been young (before). It wasn’t just *born* old!

From these four pieces of evidence, we conclude that Paraguayan Guaraní *-ma* is an *already*-like expression.

3 Change-of-state inference is an entailment for *-ma*

While both English *already* and Paraguayan Guaraní *-ma* result in a change-of-state inference with statives, it has been observed that this inference is **not an entailment for English *already*** (Ippolito 2007, Mittwoch 1993, cf. Löbner 1989).

- Indicated by the lack of contradiction in (5), where a change of state from *not speaks Guaraní* to *speaks Guaraní* is not necessary for felicity.

- (5) Context: A's husband is a speaker of Paraguayan Guaraní, his native language.
 B: Now that you're taking Guaraní classes, is your husband also learning it?
 A: He **already** speaks Guaraní. (It's his first language.)

However, Paraguayan Guaraní *-ma* instead results in a contradiction (6).

- It is infelicitous to use *-ma* when Speaker A's husband has always spoken Guaraní.
 - Thus, the change-of-state inference is an **entailment** of *-ma*.
- (6) Context: A's husband is a speaker of Paraguayan Guaraní, his native language.
 B: Now that you're taking Guaraní classes, is your husband also learning it?
 A: Ha'e o-ñe'ẽ(-#ma) avañe'ẽ. (Péva ha'e i-ñe'ẽ michĩ=guive.)
 3.PRON 3.ACT-speak(-ma) Guaraní that 3.PRON 3.POSS-language little=since
 'He (#**already**) speaks Guaraní. (That's been his language since he was little.)'

We thus propose that there are **two types of aspectual *already* cross-linguistically**:

- **Type A:** Entails a change of state.
 - Paraguayan Guaraní *-ma* (6), Bulgarian *veče* (7b)
 - **Type B:** Only implicates a change of state.
 - English *already* (5), Spanish *ya* (7a)
- (7) Context: A's husband is a speaker of Paraguayan Guaraní, his native language.
 B: Now that you're taking Guaraní classes, is your husband also learning it?
- a. A: Ya habla guaraní.
 already speak.3SG.PRS Guaraní
 'He **already** speaks Guaraní.' (Spanish)
- b. A: Toj (#veče) govori Guarani.
 3.MASC already speak.3SG.PRS Guaraní
 'He (#**already**) speaks Guaraní.' (Bulgarian)

An additional contrast between English *already* and Paraguayan Guaraní *-ma*: Despite having the semantic properties of *already*, *-ma* is **an affix** rather than an adverbial.

- We hypothesize that Type A *already*s are especially likely to grammaticalize (Olsson, 2013; Dahl & Wälchli, 2016).

4 Analysis of aspectual *already*s

How is the contrast in meaning between the two types of aspectual *already* best modeled?

- To answer this question, we start by addressing the core meaning of *already* that is shared across both types.

Analysis of aspectual *already* by Ippolito (2007):

$$(8) \quad \llbracket \text{already} \rrbracket^{g,c,w} = \lambda t \in D_i. \lambda e \in D_1. \lambda P \in D_{\langle 1, \langle i, t \rangle \rangle}: \exists t' > t [P(e)(t') = 1]. P(e)(t) = 1$$

- Under this analysis, the ‘earliness’ implication arises via the **presupposition of (8)**: There exists a time t' after t_c s.t. $P(e)$ holds of t' .

However, because the presupposition makes direct reference to the time of the eventuality, **past perfective sentences with aspectual *already*** pose a challenge:

- (9) a. It already rained.
- b. $\llbracket \text{PST}_2 \text{ already PRFV rain} \rrbracket^{g,c,w}$ is defined iff
 $\exists t' > t' [\tau(e) \subseteq t' \ \& \ \forall w' [\tau(e_1) \text{ is a time when it rains in } w']]$.
 If defined, $\llbracket \text{PST}_2 \text{ already PRFV rain} \rrbracket^{g,c,w} = 1$ iff $\exists t' [t' < t_c \ \& \ t' \subseteq g(2) \ \& \ \tau(e_1) \subseteq t' \ \& \ \forall w' [\tau(e_1) \text{ is a time when it rains in } w']]$.
- (9b) both presupposes that $\tau(e_1)$ is contained in a time t'' **and** asserts that $\tau(e_1)$ is contained in a separate, non-overlapping time t' .

In order to maintain the key insights of Ippolito’s (2007) account while also capturing the compatibility of aspectual *already* with past perfectives, we model the counterexpectation of *already* based on **Toosarvandani (2014) for *but***:

- (10) a. $\llbracket \phi \text{ but } \psi \rrbracket = \exists p [p \in \text{QUD}(\llbracket \phi \rrbracket \Rightarrow p)] \ \& \ \exists p' [p' \in \text{QUD}(\llbracket \psi \rrbracket \Rightarrow \neg p')]$. $\llbracket \phi \rrbracket \ \& \ \llbracket \psi \rrbracket$
- b. **A:** Is your cup of tea tasty?
B: It’s sweet, but bitter.
- c. (i) It’s sweet. $\Rightarrow$ The cup of tea is tasty.
 (ii) It’s bitter. $\Rightarrow$ The cup of tea is not tasty.

- Counterexpectation is formalized via the weak implications (i.e., $\Rightarrow$) in the presupposition of (10a).

Analysis of Type A versus Type B aspectual *already*s:

- All modify an AspP of type $\langle i, t \rangle$.
- Differ in their presuppositions, with identical at-issue entailments.
- **Paraguayan Guaraní -ma:** Presupposes a change of state from $\neg p$ to p , implemented via propositional BECOME operator (Stechow, 1996; Beck et al., 2001) in (12).
- **English *already*, Spanish *ya*:** No change-of-state presupposition.

- (11) a. $\llbracket \text{already}_Q^{\text{ASP}} \rrbracket^{g,c} = \lambda p. \lambda t: \exists p' [p' \in Q \ \& \ p' \Rightarrow \neg p(t) \ \& \ \exists t' [t' > t \ \& \ p' \Rightarrow p(t')]]$. $p(t)$
- b. $\llbracket \text{-ma}_Q^{\text{ASP}} \rrbracket^{g,c} = \lambda p. \lambda t: \exists p' [p' \in Q \ \& \ p' \Rightarrow \neg p(t) \ \& \ \exists t' [t' > t \ \& \ p' \Rightarrow p(t')]] \ \& \ \text{BECOME}_t(p)$.
- (12) $\text{BECOME}_t(p) := t$ is the smallest interval s.t. $\exists t' [\text{RB}(t') = \text{LB}(t) \ \& \ \neg p(t') \ \& \ \exists t'' [\text{RB}(t) = \text{LB}(t'') \ \& \ p(t'')]]$
 (Stechow, 1996; Beck et al., 2001)

Counterexpectational “earliness” implication: Arises via sensitivity to the discourse structure, following Toosarvandani (2014) on counterexpectational *but*.

(11a) and (11b) presuppose that there is an answer p' to a contextually-salient question Q , where p' **implies**:

- (i) The asserted proposition p doesn't hold at the reference time t .
- (ii) p will hold at a future time t' .

Exemplified for the present progressive sentence (13), adapted into a dialogue from (1a):

(13) **A:** When will it start raining?

B: It is already raining.

- p = It is raining.
- p' = e.g., It will start raining tonight.
- p' implies **both** that it is not raining now (i.e., $\neg p$) and that it will rain later.

Exemplified for the past perfective sentence (14):

(14) **A:** When will it start raining?

B: It already rained earlier.

- p = It rained earlier.
- p' = e.g., It will start raining tonight.
- p' implies **both** that it did not rain earlier (i.e., $\neg p$) and that it will rain later.

5 Analysis of marginal *already*s

In addition to their aspectual uses, *already/ya* and *-ma/veče* also have a **marginal use** (15).

- Occurs when **the QUD is a multiple-*wh* question**, pairing individuals with degrees.
- Signals that the predicate holds of the individual currently being discussed, but not of another salient individual (Michaelis, 1993; Ippolito, 2007).

(15) a. *Context: You're at a store, telling a friend how expensive each watch is.*

Reloj peteĩha nda-hepyi-ri. Reloj mokõiha hepy-**ma**. Ha reloj mbohapyha
 watch first NEG-expensive-NEG watch second expensive-*ma* CONJ watch third
 hepy-etereí-**ma**.
 expensive-very-*ma*

'The first watch isn't expensive. The second watch is **already** expensive. And, the third watch is **already** very expensive.'

b. *Context: You're showing your friend a map, discussing the geography of South America.*

Formósa o-ĩme Arhentína=pe. Ha uperire ikatu ja-hecha Asunsiõ katu
 Formosa 3.ACT-be.located Argentina=LOC CONJ after can 1PL.ACT-see Asunción just
 oĩ-**ma** Paraguái=pe.
 there.is-**ma** Paraguay=LOC

'Formosa is in Argentina. And then we can see Asunción is **already** in Paraguay.'

Analysis of marginal *already*s:

- Attach lower than Asp and modify an $\langle e, t \rangle$ predicate.
- **Discourse-sensitive, like their aspectual counterparts**, presupposing an answer p' to a contextually-salient question Q .
- Require Q to be a subquestion within a strategy of inquiry S for answering the QUD (Roberts, 1996/2012).

$$(16) \quad \llbracket \text{already}_{Q, S_{\leq \delta}}^{\text{MAR}} \rrbracket^{g,c} = \llbracket -\text{ma}_{Q, S_{\leq \delta}}^{\text{MAR}} \rrbracket^{g,c} = \lambda P. \lambda x: \exists p' [p' \in Q \ \& \ Q \in S_{\leq \delta} \ \& \ p' \Rightarrow \neg P(x) \ \& \ \exists y \ \exists Q' [y \neq x \ \& \ Q' \in S_{\leq \delta} \ \& \ Q' <_{\delta} Q \ \& \ \exists p'' [p'' \in Q' \ \& \ p'' \Rightarrow \neg P(y)]]]. P(x)$$

A **scalar ordering** $\leq_{\delta}$ is imposed on S :

- Based on the order of individuals along the dimension δ determined by the predicate.
- Analogous to the ordering of alternatives for scalar *only* / *even* (Beaver & Clark 2009).
- In (16), Q is higher on $S_{\leq \delta}$ than another subquestion Q' .

Exemplified for (17), repeated from (15a):

(17) *Context: You're at a store, telling a friend how expensive each watch is.*

Reloj peteĩha nda-hepyĩ-ri. Reloj mokõiha hepy-**ma**. Ha reloj mbohapyha
 watch first NEG-expensive-NEG watch second expensive-*ma* CONJ watch third
 hepy-etereĩ-**ma**.
 expensive-very-*ma*

'The first watch isn't expensive. The second watch is **already** expensive. And, the third watch is **already** very expensive.'

- QUD = Which watch is how expensive?
- Q = How expensive is the second watch? • Q' = How expensive is the first watch?
- p' = The second watch is not expensive. • p'' = The first watch is not expensive.
- p' implies, naturally, that the second watch is not expensive (i.e., $\neg P(x)$), **and** p'' implies that the first watch is not expensive (i.e., $\neg P(y)$).

6 Conclusion

Paraguayan Guaraní *-ma* is not a perfect nor a perfective marker, but instead, **a type of *already***.

However, Paraguayan Guaraní *-ma* is **not an adverbial** like English *already*, but is instead an affix; an element in the functional spine (similar to Medumba *yǎ*, Javanese *wis*).

In this talk, we have demonstrated that meaning-wise, there are **two types of aspectual *already*s** cross-linguistically:

- Those that presuppose a change-of-state (Paraguayan Guaraní *-ma*, Bulgarian *veče*) versus those that do not (English *already*, Spanish *ya*).

Despite the contrast in meaning between the two aspectual *already*s, both types retain their **marginal uses**.

- This proposal captures both the aspectual and marginal cases via a discourse-sensitive account of the semantics of *already*.

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